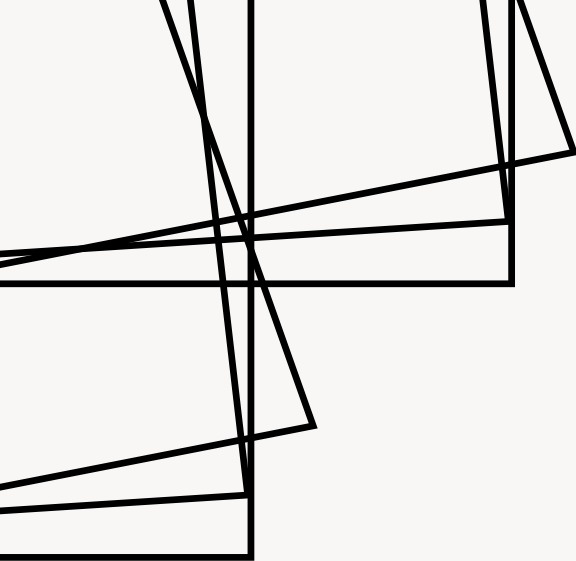


MINESWEEPER

**BY
ASSIUT ROBOTICS
TEAM**



2.ELECTRICAL SECTION



ELECTRICAL SYSTEM OVERVIEW

Our robot uses a dual-controller architecture consisting of:

- Raspberry Pi 4 (8GB) for high-level processing.
- Arduino Mega for low-level hardware control.

The Raspberry Pi is responsible for:

- Computer Vision
- Localization
- Mine Mapping
- Running ROS 2

The Arduino is responsible for:

- Motor Control
- Reading Sensors
- Executing real-time hardware tasks

Both controllers communicate through Serial Communication, ensuring fast and reliable data exchange between processing and control units.



- **Hardware Components**
- **The robot contains several electronic components working together:**
- **Raspberry Pi 4**
- **Arduino Mega**
- **DC Motors**
- **Motor Drivers**
- **Wheel Encoders**
- **IMU Sensor**
- **Metal Detector Circuit**
- **Proximity Sensors**
- **HD Camera**
- **Warning LED**
- **Buzzer**
- **Each component has a specific role that contributes to autonomous navigation and mine detection.**

MAIN ELECTRONIC COMPONENTS

SENSORS AND CONTROL

Sensors Integration

The electrical system combines multiple sensors to improve robot performance.

- Wheel Encoders measure traveled distance.
- IMU provides robot orientation and heading.
- Metal Detector identifies buried mines.
- Proximity Sensors detect surface mines.
- HD Camera recognizes arena coordinates.

Whenever a mine is detected:

- Warning LED turns ON.
- Buzzer sounds automatically.
- Detection data is sent to the controller for processing

PCB DESIGN

Custom PCB Design

TO SIMPLIFY WIRING AND IMPROVE MAINTENANCE, THE ELECTRONICS ARE DIVIDED INTO THREE CUSTOM PCB MODULES.

1. Metal Detector Module

- Metal Detector
- Five Proximity Sensors
- Status LEDs

2. LIFT MECHANISM MODULE CONTROLS:

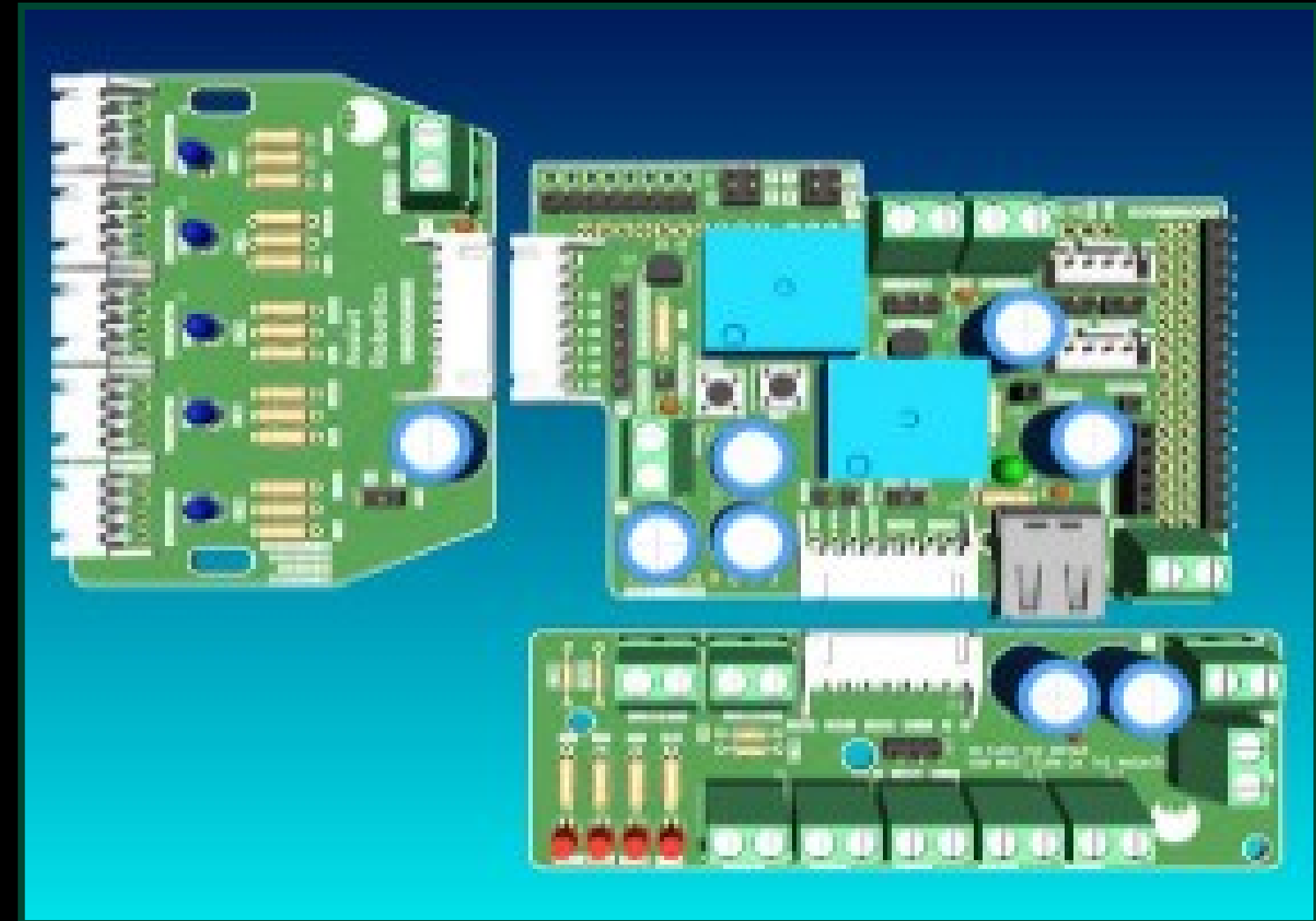
- ELECTROMagnets
- DC Motor
- Limit Switches

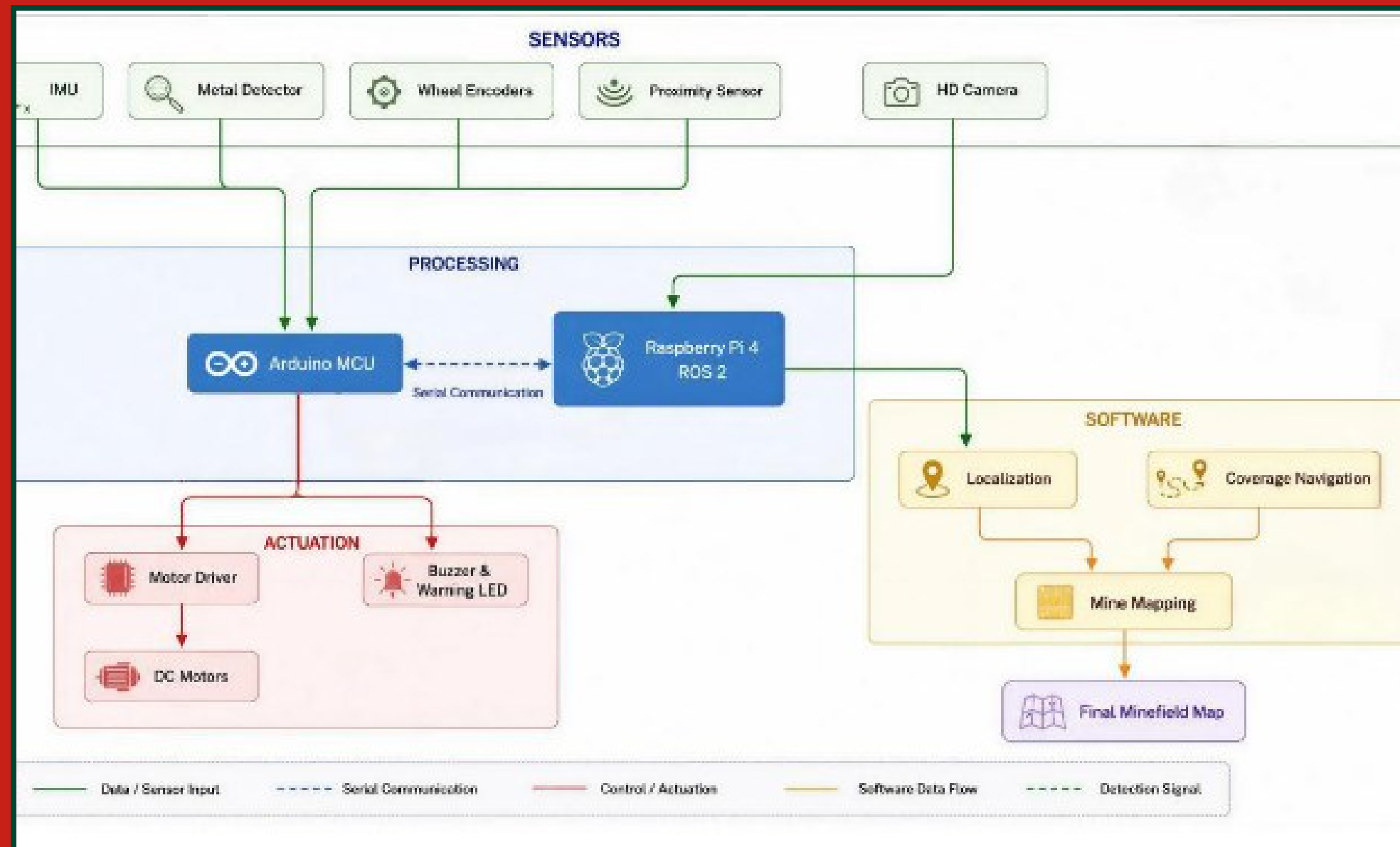
3. Centralized Module

MOUNTED DIRECTLY ON THE ARDUINO MEGA AS A SHIELD.
RESPONSIBLE FOR:

- POWER DISTRIBUTION
- SENSOR Connections
- COMMUNlcation with Raspberry Pi

This modular design reduces cable complexity and makes troubleshooting easier





Software Architecture

The robot is controlled using ROS 2.
DeDICATED ROS NODES MANAGE:

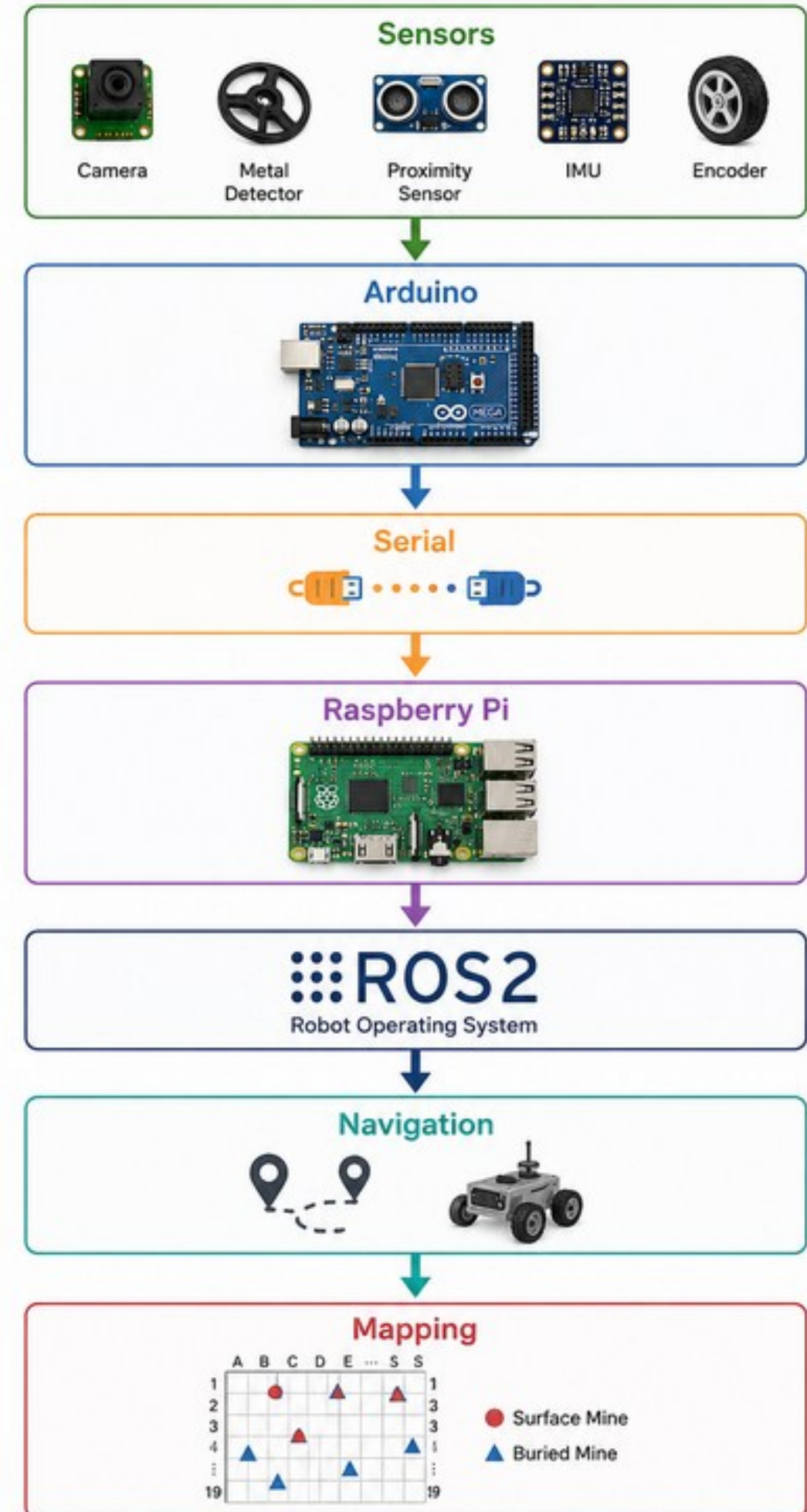
- VISION PROCESSING
- LOCALIZATION
- MINE DETECTION
- NAVIGATION
- MINEFIELD MAPPING

THE ROBOT MOVES AUTONOMOUSLY USING A LAWNMOWER (ZIGZAG) COVERAGE STRATEGY, SCANNING THE ENTIRE FIELD WHILE GENERATING THE MINE MAP AUTOMATICALLY.

SOFTWARE & CONTROL ARCHITECTURE

SYSTEM WORKFLOW

This workflow illustrates how sensor data flows through the controllers and software to produce autonomous robot movement and mapping.



An abstract geometric frame composed of several overlapping black lines. It includes a central rectangle and several other lines that intersect it at various angles, creating a complex, layered effect. The lines are black and have a consistent thickness.

THANK YOU